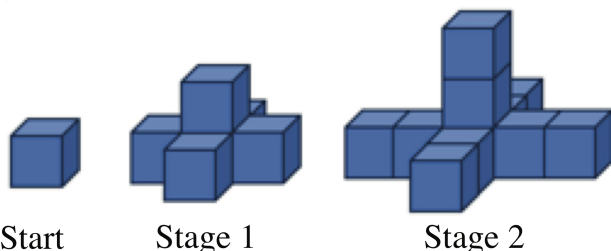


FSJM – SEMI-FINAL- 15 MARCH 2025

Information and results at <http://www.fsjm.ch/>

START for ALL PARTICIPANTS

1. Evolution (coefficient 1)



Lucas is making a model using small cubes.

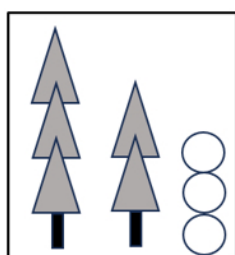
At each stage, he adds a cube to each end of his model, as shown in the diagram.

How many small cubes will be included in his model when he reaches stage 4?

2. Sticker Pictures (coefficient 2)

The teacher suggests making this picture with grey, white and black stickers.

In the box of stickers are:

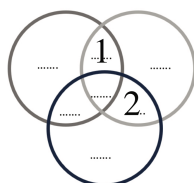


- 10 black stickers;
- 15 white stickers;
- 20 grey stickers.

How many pictures can we make at most?

3. Sum Circles (coefficient 3)

This figure is made of 3 circles.



Place the numbers 3, 4, 5, 6 and 7 in the empty spaces.

Be careful: when you add the four numbers in any circle, you must obtain 14.

4. Sudoku Stickers (coefficient 4)

Lily wants to place white stickers ○ and black stickers ● on the squares of this grid.

She has already placed 3 black stickers and 3 white stickers.

Each square must contain one, two or three stickers of a single colour;
in each row or column, no square has the same number of stickers as another;

the number at the beginning of a row or column indicates the total number of black stickers in that row or column.

Place the black stickers ● and white stickers ○ in Lily's grid.

5. Odds and Evens (coefficient 5)

March 15, 2025
In this box, there are:
... even digits, and ... odd digits

Complete the sentence written in the box with numbers written in figures so that it is true.

Note: the even numbers are 0, 2, 4, 6 and 8.

END for CE PARTICIPANTS

6. A B C (coefficient 6)

Clara must practise her clarinet to improve her fingering.

She must play the three notes A, B, C in groups of 4, (such as A-B-C-B) with the following three conditions:

- in each group of 4, there are the 3 notes,
- two notes that follow each other must be different,
- the first and the last must not be the same.

How many different groups can she form?

7. Stirring Subtraction (coefficient 7)

$$\begin{array}{r} \text{DAME} \\ - \text{EDAM} \\ \hline = 2025 \end{array}$$

In this subtraction, the same letter always replaces the same digit, and two different letters always replace two different digits. No letter replaces a 0.

What is DAME worth?

8. Nine Pawns (coefficient 8)

8	1	6
3	5	7
4	9	2

Nine pawns are placed on this grid. Matilda wants to remove 3 or 4 of the grey pawns, but leaving the 5, her favourite number. The total of the numbers on the pawns remaining on the grid (including the 5) must equal 25.

How many different arrangements of the pieces remaining on the grid are there?

END for CM PARTICIPANTS

Problems 9 to 18: beware! For a problem to be completely solved, you must give both the number of solutions, and give the solution if there is only one, or give any two correct solutions if there are more than one. For all problems that may have more than one solution, there is space for two answers on the answer sheet (but there may still be just one solution).

9. Matilda's Fraction (coefficient 9)

Matilda wrote a fraction with a numerator strictly greater than 25 and a denominator strictly smaller than 2×25 . This fraction is equal to $\frac{2}{3}$.

What is the fraction Matilda wrote?

10. Circular Puzzle (coefficient 10)

A circular puzzle consists of a central piece and rings of pieces around it. The innermost ring has a certain number of pieces (at least 5), and each surrounding ring has more pieces than the adjacent enclosed one (at least 3 more). We also know that the difference in the number of pieces between two consecutive rings is always the same.

How many rings are there in a puzzle with 625 pieces, knowing that there are 25 pieces if we only count the central piece and the first two rings?

11. A Means to a Mean (coef. 11)

Around a circle are drawn nine straight lines that are tangent to it, to obtain an irregular polygon with nine vertices.

What is the average of the measurements in degrees of the nine angles?

END for C1 PARTICIPANTS

12. Square and Trapeziums (coef. 12)

On the extension [AB] of one of the sides of a square ABCD of side 74 cm, we place the point M such that $BM = 13$ cm. A line is drawn through M that divides the square into two trapeziums of the same area, cutting [BC] at point E.

What is the length of BE in cm?

Give the answer to the nearest 0.01 cm.

13. Strange Sum of Squares (cf. 13)

$$10^2 + 1^2 = 101$$

Find two two-digit positive integers A and B such that:

$$A^2 + B^2 = 100A + B.$$

Give the value of $100A + B$.

14. Square of Squares (coef. 14)

	2		
2	0	2	5
	2		
	5		

Fill in the empty boxes of this square so that:

- the numbers in each row, read from left to right, are all squares of whole numbers;
- the numbers in each column, read from top to bottom, are all squares of whole numbers.

What will be the number in the first row?

END for C2 PARTICIPANTS

15. Outdo Yourself! (coefficient 15)

Find two positive integers knowing that the first exceeds quintuple the second by 31, and that the product of the two numbers exceeds their sum by 2249.

16. Properties of my Triangle (coefficient 16)

The sides BA, AC and BC of a triangle are, in this order, in strictly increasing arithmetic progression with a common difference of " r ".

Moreover:

$$BA \times BC^2 = AC^3 + 11r^3.$$

Calculate BC in cm knowing that the area of the triangle is 600 cm².

Give the answer to the nearest 0.01 cm.

END for L1, GP PARTICIPANTS

17. Number Boxes (coefficient 17)

Boxes are numbered 1, 2, 3, etc. to infinity. Each box contains a positive integer. For an integer n , we denote $s(n)$ as the sum of its digits.

We know that:

- for all n , the number in box $n+s(n)+s(n+s(n))$ is equal to the sum of the numbers in boxes n and $n+s(n)$;
- boxes 1 and 2 contain strictly positive integers;
- box 7 contains the number 13.

What integer does box 64 contain?

18. Brahmagupta's Triangles

(coefficient 18)

The triangle with sides of length 3, 4 and 5 is remarkable. The lengths of its sides are consecutive integers, and its area is also an integer. The mathematician Brahmagupta found another triangle with these properties.

What is its perimeter, knowing that it is less than 2025?

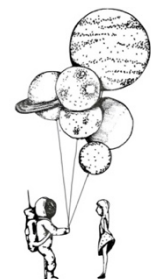
END for L2, HC PARTICIPANTS

The Swiss Federation of Mathematical Games warmly thanks its sponsors for their help in organizing this event



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